Engineering Linear Algebra

LU Decomposition using Elementary Matrices E

Kacper Cichy

Monday

Learning Objective for Today

Understand the process of LU factorization using elementary matrices E and optimize the use of space on the worksheet.

Theory: LU Decomposition and Elementary Matrices

During Gaussian elimination, we apply elementary row operations to transform matrix A into an upper triangular matrix U. Each such operation can be written as multiplication by an elementary matrix E.

- **Elementary matrix E_{ij}:** An identity matrix in which the value $-m$ is placed in the i-th row and j-th column, where m is the multiplier used to zero out a given element.
- **Elimination process:** We write the sequence of multiplications $E_k \cdots E_2 E_1 A = U$.
- **Construction of L:** If we move the E matrices to the right side, we obtain $A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U$. Thus, $L = E_1^{-1} E_2^{-1} \cdots E_k^{-1}$.
- **The magic of inverses:** The inverse of an elementary matrix E_{ij}^{-1} is simply the same matrix, but with the **sign of the multiplier changed** (from $-m$ to m). Moreover, their product consists of inserting these multipliers with changed signs into the appropriate places of the identity matrix.

Step-by-Step Example (Elementary Matrices)

For the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 1 & 0 \\ -2 & 2 & 1 \end{bmatrix}$, we find the LU decomposition.

1. **Column 1:** We zero out the elements below $a_{11} = 2$.

- Row 2: subtract $2 \times R_1$. Matrix $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
- Row 3: add $1 \times R_1$ (subtract $-1 \times R_1$). Matrix $E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$.

After performing $E_{31} E_{21} A$, we get the form $\begin{bmatrix} 2 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 3 & 2 \end{bmatrix}$.

2. **Column 2:** We zero out the element below $a_{22} = -1$.

- Row 3: add $3 \times R_2$ (subtract $-3 \times R_2$). Matrix $E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$.

After performing $E_{32}(E_{31}E_{21}A)$, we obtain $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & -4 \end{bmatrix}$.

3. Creating matrix L: $L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$. We change the signs below the diagonal:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix}$$

Training Set

Problems to be solved on a separate sheet of paper. Scheme for each system $Ax = b$: (1) Determine E, L, U, (2) Solve $Ly = b$, (3) Solve $Ux = y$.

Size 2x2

- **Problem 1:** $\begin{cases} 4x_1 + 3x_2 = 10 \\ 8x_1 + 5x_2 = 18 \end{cases}$

Size 3x3

- **Problem 2:** $\begin{cases} 3x_1 - 6x_2 + 3x_3 = -3 \\ 2x_1 + 0x_2 + 6x_3 = -22 \\ -4x_1 + 7x_2 + 4x_3 = 3 \end{cases}$

- **Problem 3:** $\begin{cases} 1x_1 + 2x_2 + 4x_3 = 3 \\ 3x_1 + 8x_2 + 14x_3 = 13 \\ 2x_1 + 6x_2 + 13x_3 = 4 \end{cases}$

Size 4x4

- **Problem 4:** $\begin{cases} 2x_1 + 1x_2 + 0x_3 + 0x_4 = 1 \\ 4x_1 + 5x_2 + 1x_3 + 0x_4 = 1 \\ 2x_1 - 2x_2 - 1x_3 + 1x_4 = 1 \\ 0x_1 + 6x_2 + 0x_3 - 2x_4 = -4 \end{cases}$

- **Problem 5:** $\begin{cases} 1x_1 + 2x_2 + 0x_3 + 1x_4 = 1 \\ 1x_1 + 1x_2 + 1x_3 + 3x_4 = 0 \\ 2x_1 + 1x_2 + 5x_3 + 7x_4 = 2 \\ -1x_1 - 3x_2 + 5x_3 + 0x_4 = 3 \end{cases}$

Answer Key (Resulting vectors x):
Prob. 1: $x = [1, 2]^T$
Prob. 2: $x = [-1, 2, -4]^T$
Prob. 3: $x = [3, 4, -2]^T$
Prob. 4: $x = [1, -1, 2, -1]^T$
Prob. 5: $x = [2, 0, 1, -1]^T$